INTERNAL HOM FROM A PERFECT OBJECT INTO A BOUNDED FLAT SUPPORTED COMPLEX IS PERFECT AND COMMUTES WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is a complete approximation chain: affine reduction, descent of X and of the datum to some R_i including flatness and proper support and a perfect E_i, then the Noetherian case plus the proved base change property yields the result

Editorial note. The label lemma-base-change-RHom-perfect names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- base change for K = Rf_*RSheafHom(E, G) with E perfect and G bounded, finitely presented, flat over Y with support proper over Y (lemma-base-change-RHom of the same chapter)
- perfectness of Rf_*RSheafHom(E, G) when Y is Noetherian (lemma-ext-perfect of the same chapter)
- Noetherian approximation: descent of finite presentation for spaces and modules, of flatness, and of proper support along R = colim R_i, from Limits of Spaces
- descent of perfectness of E along the Noetherian approximation (lemma-descend-perfect of the same chapter)
- the perfectness computation of Rf_*RSheafHom over a Noetherian base (lemma-compute-ext-perfect of the same chapter)

Source reference. Original reference 08IE: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5675–5675. Prerequisite pointer invoked at proof line(s) 6599; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08ID: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5946–5946. Prerequisite pointer invoked at proof line(s) 6627; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3288–3288. Prerequisite pointer invoked at proof line(s) 6624; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DJR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5820–5820. Prerequisite pointer invoked at proof line(s) 6601; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07SK: sources/stacks-chow/tags.txt, source lines 9905–9905. Prerequisite pointer invoked at proof line(s) 6609; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08K0: sources/stacks-chow/tags.txt, source lines 10868–10868. Prerequisite pointer invoked at proof line(s) 6620; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07V7: sources/stacks-chow/tags.txt, source lines 9997–9997. Prerequisite pointer invoked at proof line(s) 6613; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08K2: sources/stacks-chow/tags.txt, source lines 10870–10870. Prerequisite pointer invoked at proof line(s) 6623; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of finite presentation between algebraic spaces over S. Let $E \in D(\mathcal{O}_X)$ be a perfect object. Let $\mathcal{G}^\bullet$ be a bounded complex of finitely presented $\mathcal{O}_X$-modules, flat over Y, with support proper over Y. Then*

$$K = Rf_*R\mathcal{H}om(E, \mathcal{G}^\bullet)$$

is a perfect object of $D(\mathcal{O}_Y)$ and its formation commutes with arbitrary base change.

Proof. The statement on base change is Lemma 08IE. Thus it suffices to show that K is a perfect object. If Y is Noetherian, then this follows from Lemma 0DJR. We will reduce to this case by Noetherian approximation. We encourage the reader to skip the rest of this proof.

The question is local on Y, hence we may assume Y is affine. Say $Y = \mathrm{Spec}(R)$. We write $R = \mathrm{colim}\, R_i$ as a filtered colimit of Noetherian rings R_i. By Limits of Spaces, Lemma 07SK there exists an i and an algebraic space X_i of finite presentation over R_i whose base change to R is X. By Limits of Spaces, Lemma 07V7 we may assume after increasing i, that there exists a bounded complex of finitely presented $\mathcal{O}_{X_i}$-module $\mathcal{G}_i^\bullet$ whose pullback to X is $\mathcal{G}$. After increasing i we may assume $\mathcal{G}_i^n$ is flat over R_i, see Limits of Spaces, Lemma 08K0. After increasing i we may assume the support of $\mathcal{G}_i^n$ is proper over R_i, see Limits of Spaces, Lemma 08K2. Finally, by Lemma 08HG we may, after increasing i, assume there exists a perfect object E_i of $D(\mathcal{O}_{X_i})$ whose pullback to X is E. Applying Lemma 08ID to $X_i \to \mathrm{Spec}(R_i)$, E_i, $\mathcal{G}_i^\bullet$ and using the base change property already shown we obtain the result. □